PROVISIONAL PATENT APPLICATION

Title: SHIMMERS: High-Frequency Digital Pulse Generation System Using Serializer-Based Direct Synthesis, Shared-Root Clock Management, and CRC64-Gated Integrity Delivery

---

COVER SHEET

Application Type: Provisional Patent Application (35 U.S.C. § 111(b))

Title of Invention: High-Frequency Digital Pulse Generation System Using Serializer-Based Direct Synthesis with Adaptive Mixed-Mode Clock Management, MIDI-Mapped Bijective Frequency Encoding, and CRC64-Integrity-Gated Delivery

Inventor: Scott Slater
700 E. 8th St.
Unit 14E
Kansas City, MO 64106-5400
United States Citizen

Filing Date: May 6, 2026

Docket / Reference: SES-001-SHIMMERS

Correspondence Address:
700 E. 8th St.
Unit 14E
Kansas City, MO 64106-5400

FIELD OF THE INVENTION

This invention relates to digital electronic systems for high-frequency pulse generation and frequency synthesis on reconfigurable logic devices (FPGA), and more particularly to a novel architecture combining hardware-native output serializers, Mixed-Mode Clock Managers (MMCM), phase accumulator-based frequency encoding, sensor-driven adaptive regulation, and integrity-sealed delivery to achieve pulse generation rates exceeding 600 MHz with deterministic timing, minimal jitter, and dynamic environmental compensation — capabilities that fundamentally exceed the performance ceiling of prior art Direct Digital Synthesis (DDS) systems.

BACKGROUND

1.1 Limitations of Prior Art Direct Digital Synthesis

Traditional Direct Digital Synthesis (DDS) systems generate periodic waveforms by accumulating a phase increment register and using the resulting phase value to index a lookup table (LUT) or Read-Only Memory (ROM) that stores pre-computed amplitude values. This approach suffers from several fundamental limitations:

Phase Truncation Error: As the required output frequency increases, the phase accumulator must be truncated to address a finite-size LUT. This truncation introduces spurious spectral components that degrade the Spurious-Free Dynamic Range (SFDR), typically limiting performance to -70 dBc to -88 dBc.

ROM Size Scaling: Higher frequency resolution requires exponentially larger lookup tables. A 32-bit phase accumulator addressing a 512-entry sine LUT wastes 23 address bits, while increasing LUT size to match accumulator width is resource-prohibitive.

Clock Rate Ceiling: Conventional FPGA fabric logic cannot sustain operation at multi-hundred MHz rates due to routing congestion, hold time violations, and clock skew across distributed logic structures.

Waveform Reload Latency: When pulse parameters change, ROM-based systems require waveform memory to be reloaded, introducing phase discontinuities and transition latency that is unacceptable in real-time encoding applications.

Trigger Uncertainty: ROM-based DDS systems suffer from up to one full clock cycle of trigger uncertainty, limiting the precision of frequency-keyed data encoding applications.

Thermal and Voltage Sensitivity: Fixed-configuration clock synthesizers are susceptible to frequency drift under thermal load and supply voltage variation, with no mechanism for real-time compensation.

1.2 The OSERDESE3 Clock Skew Problem

High-speed FPGA output serializer primitives (e.g., OSERDESE3 on UltraScale+) require that their high-speed clock (CLK) and their parallel-load divided clock (CLKDIV) arrive within a strict maximum skew specification — 0.193 ns at the Fast process corner on UltraScale+ devices.

When CLK and CLKDIV are derived from independent MMCM outputs routed through separate clock buffers placed at different physical locations on the device, inter-tree routing skew routinely exceeds this specification. No prior art is known that solves this skew problem by deriving CLKDIV from the same pre-buffer clock source as CLK, ensuring both signals share an identical clock tree root.

1.3 Summary of the Problem

What is needed is a pulse generation system that: (a) exceeds 600 MHz sustained output rate on commercially available FPGA hardware; (b) eliminates ROM-based waveform lookup in favor of memoryless serialization; (c) provides instantaneous frequency switching without phase discontinuity; (d) compensates in real-time for thermal drift and supply variation; (e) guarantees lossless data integrity through cryptographic sealing; (f) operates deterministically with no stalls permitted in the pulse path; and (g) is portable across multiple FPGA device families without RTL modification. The Shimmers invention addresses all seven requirements.

SUMMARY OF THE INVENTION

Shimmers is a high-frequency digital pulse generation system implemented as synthesizable VHDL targeting reconfigurable hardware (FPGA) that replaces ROM-based waveform synthesis with a hardware-native OSERDESE3 output serializer driven by a Mixed-Mode Clock Manager. The system encodes arbitrary 7-bit data symbols as frequency-modulated pulse streams using a bijective MIDI-mapped lookup table, seals each pulse frame with a CRC64-ECMA-182 integrity check, and delivers the sealed stream downstream through a Conductor finite state machine that gates delivery on seal validity.

The central clock innovation is a novel topology in which the serializer's divided clock (CLKDIV = 150 MHz) is derived from the pre-buffer high-speed clock signal (clk_fast_unbuf) via a BUFGCE_DIV primitive rather than from a separate MMCM output. This ensures CLK and CLKDIV share an identical clock tree distribution root, eliminating the inter-tree routing skew that causes OSERDESE3 Max Skew violations in prior art implementations. On UltraScale+, this reduces worst-case CLK-to-CLKDIV skew from 0.275 ns (violation) to within 0.066 ns (compliant), a 76% reduction.

A second innovation is the bijective MIDI-mapped frequency LUT: a 128-entry synchronous ROM mapping each 7-bit input symbol to a unique 32-bit phase increment word. The bijective property guarantees that every input symbol maps to a distinct output frequency, making the mapping losslessly invertible by a downstream Goertzel-based decoder. The Huffman pre-encoder enforces MSB=0 on all output bytes, guaranteeing MIDI note range validity (0–127) before the LUT is accessed.

A third innovation is the CRC64-ECMA-182 hard delivery gate: no pulse frame may be promoted to a downstream lane unless the CRC64 seal module has asserted seal_intact. This is not advisory — it is a combinational interlock in the Conductor FSM RTL.

A fourth innovation is the Sysmon feedback loop: on-chip junction temperature and core supply voltage are continuously measured by the XADC/Sysmon hard IP, converted to physical measurements, and fed to an oscillator modifier pipeline enabling real-time frequency compensation without system halt or bitstream reload.

When combined with the Eidolon Dragoncache IPC engine (SES-002) and the SPECTRE frequency-domain transceiver (SES-003), Shimmers forms the oscillator and frequency-source tier of the Shimmers · Eidolon · SPECTRE (SES) integrated system.

BRIEF DESCRIPTION OF DRAWINGS

(Informal drawings — formal drawings to accompany non-provisional application)

Figure 1 — Top-Level System Block Diagram Data flow from input byte through Huffman encoder, frequency LUT, phase accumulator, pattern encoder, OSERDESE3 serializer, OBUFDS differential output. Clock tree shown separately (Figure 2).

(Clock detail: see FIG. 2)

Figure 2 — Clock Domain Architecture IBUFDS differential input → MMCME4_ADV (VCO=1200 MHz) → BUFG for clk_fast (600 MHz) and BUFGCE_DIV(÷4) for clk_div (150 MHz) sharing same clk_fast_unbuf root → BUFG for clk_100m (100 MHz). BUFGCE_DIV co-location with bufg_fast shown. Prior art independent BUFG path shown with skew annotation (0.275 ns > 0.193 ns limit).

Figure 3 — Frequency LUT Mapping 128-entry table: 7-bit MIDI note index (0–127) → 32-bit phase increment word → target frequency in Hz (8.18 Hz to 12,543.85 Hz). Bijective property annotated: each input maps to exactly one output, and each output maps back to exactly one input.

Figure 4 — Phase Accumulator and Pattern Encoder Signal Flow 32-bit phase_inc → 32-bit accumulator (advances each clk_100m cycle, wraps at 2^32) → 8-bit parallel pattern → OSERDESE3 D[7:0] input. Valid gating from huff_encoded_valid shown.

Figure 5 — OSERDESE3 Configuration Detail DATA_RATE_OQ=DDR, DATA_WIDTH=8, CLK=600 MHz, CLKDIV=150 MHz (from BUFGCE_DIV), D[7:0] from pattern encoder, OQ serial

output → OBUFDS → pulse_out_p/n differential pair. I/O standard: DIFF_HSTL_I_12 for 1.2V HP bank.

Figure 6 — CRC64 Seal and Conductor FSM Interaction CRC64 module receives frame data, asserts seal_intact. Conductor FSM state: ST_SEAL_WAIT → ST_PROMOTE (only when seal_intact='1'). Hard gate shown as combinational AND in FSM transition logic.

Figure 7 — Sysmon Feedback Loop XADC/Sysmon hard IP → temperature code (T = CODE×503.975/4096 − 273.15) and voltage code (V = CODE×3.0/4096) → instruction word → oscillator modifier pipeline → phase_inc adjustment. xadc_alert → Conductor FSM lane_switch path shown separately.

Figure 8 — Multi-Platform Validation Table Nine device families: Spartan-7 through Versal VPK180. WNS, WHS, WPWS all positive. Validation dates May 2026. SHA-256 lossless verification column.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT

4.1 System Overview

The Shimmers pulse generation system operates as the oscillator engine of the larger SES frequency-domain data encoding and transmission architecture. It receives input data bytes from upstream processing stages, translates each byte to a corresponding frequency via a bijective mapping function, generates a high-speed pulse stream encoding that frequency, seals each frame with a 64-bit cryptographic integrity check, and delivers the pulse stream to downstream consumers via a deterministic, stall-free pipeline.

The system is implemented in synthesizable VHDL-2008 strict and has been validated on the following platforms with zero timing violations:

Platform	**Device**	**WNS (ns)**	**Validation Date**
Alveo U250	xcu250-figd2104-2L-e	+0.685	May 4, 2026
Alveo U200	xcu200-fsgd2104-2-e	validated	April 29, 2026

Platform	Device	WNS (ns)	Validation Date
Versal VPK180	xcvp1802-lsvc4072-2MP-e-S	validated	April 25, 2026
Virtex VU47P	xcvu47p-fsvh2892-2-e	+1.254	May 1, 2026
Virtex VU9P	xcvu9p-flgb2104-2-e	validated	May 2, 2026
Kintex-7	xc7k325t	validated	April 29, 2026
Artix-7	xc7a100tcsg324-1	validated	April 29, 2026
Spartan-7	xc7s50csga324-1	+11.508	May 2, 2026
Alveo V80	xcv80	PDI validated	April 30, 2026

4.2 Novel Contribution 1 — Shared-Root BUFGCE_DIV Clock Topology

The Shimmers system derives all operating clocks from a single 300 MHz differential reference input using a MMCME4_ADV. The MMCM is configured with CLKFBOUT_MULT_F=4.0, DIVCLK_DIVIDE=1, producing VCO=1200 MHz (within the UltraScale+ valid range of 800–1600 MHz). CLKOUT0_DIVIDE_F=2.0 produces clk_fast_unbuf at 600 MHz. CLKOUT2_DIVIDE=12 produces clk_100m_unbuf at 100 MHz.

The critical innovation: The serializer control clock (clk_div = 150 MHz) is NOT derived from MMCM CLKOUT1 through an independent BUFG. Instead, it is derived from clk_fast_unbuf (the pre-buffer 600 MHz signal) via a BUFGCE_DIV primitive with BUFGCE_DIVIDE=4. This means:

- clk_fast travels: clk_fast_unbuf → BUFG (bufg_fast) → clk_fast_out
- clk_div travels: clk_fast_unbuf → BUFGCE_DIV(÷4) → clk_div_out

Both paths originate from the identical node (clk_fast_unbuf). The OSERDESE3's CLK and CLKDIV inputs see signals that share the same clock tree spine from the same MMCM output, guaranteeing sub-nanosecond skew by construction.

In the prior art implementation (MMCM CLKOUT1 → independent BUFG for CLKDIV), Vivado placed bufg_fast at BUFGCE_X0Y98 and bufg_div at BUFGCE_X0Y115 — 17 physical clock buffer sites apart. The resulting inter-tree routing skew was 0.275 ns, exceeding the OSERDESE3 Max Skew specification of 0.193 ns and producing a -0.082 ns WPWS violation. The BUFGCE_DIV topology eliminated this violation entirely, with WPWS improving from -0.082 ns to +0.066 ns.

4.3 Novel Contribution 2 — Bijective MIDI-Mapped Frequency LUT

The frequency_lut module implements a 128-entry synchronous ROM mapping each 7-bit input symbol (MIDI note 0–127) to a 32-bit phase increment word. The phase increment encodes target frequency F as:

$$phase\_inc = F \times 2^{32} / F\_clock$$

where F_clock = 100 MHz. The mapping is strictly bijective: each of the 128 MIDI note frequencies (8.176 Hz to 12,543.85 Hz, equal-temperament spacing) maps to exactly one phase increment value, and each phase increment maps to exactly one MIDI note. This invertibility is the prerequisite for lossless data recovery by the downstream Goertzel decoder.

The lookup is single-cycle latency: phase_inc_out is valid one clock cycle after byte_in is presented. Throughput is one lookup per 100 MHz cycle, giving a maximum symbol rate of 100 million symbols per second — well above the 96 kHz Goertzel window rate required by SPECTRE.

4.4 Novel Contribution 3 — Huffman Pre-Encoder with MSB=0 Invariant

The huff_encoder module implements a static-ROM Huffman entropy encoder that processes input bytes from the UART intake FIFO prior to frequency lookup. The encoder indexes a 256-entry code table by input byte value, accumulates variable-length output codes in a 16-bit shift register, and emits complete 8-bit output bytes as the accumulator fills.

The MSB=0 invariant: The encoder enforces that the most significant bit of every output byte is zero. This guarantees that the output byte value is in the range 0x00–0x7F — a valid MIDI note index — before it reaches the frequency_lut. Without this invariant, input bytes with values 0x80–0xFF would produce out-of-range LUT indices. The huff_encoder eliminates this failure mode structurally.

The huff_encoded_valid signal directly gates phase_inc_valid in the frequency_pattern_encoder, ensuring the phase accumulator advances only when a valid encoded byte is available. This maintains cycle-accurate synchronization between entropy encoding and frequency encoding.

4.5 Novel Contribution 4 — Phase Accumulator and Pattern Encoder

The frequency_pattern_encoder receives the 32-bit phase increment word and implements a digital phase accumulator advancing by phase_inc each 100 MHz clock cycle, wrapping at 2^32. The instantaneous 32-bit phase value is mapped to an 8-bit parallel output pattern representing one OSERDESE3 serialization unit — 8 consecutive output bits.

The 8-bit pattern is presented to OSERDESE3 D[7:0] each clk_div (150 MHz) cycle. Because OSERDESE3 serializes 8 bits per clk_div cycle at 4× the clk_div rate (DDR on 600 MHz CLK), the effective output bit rate is 8 × 150 MHz = 1,200 Mbps, corresponding to 600 MHz effective pulse transitions.

The encoder is gated by a foresight_gate signal from the downstream Dragoncache hub, ensuring pulses are only emitted when all downstream components have confirmed readiness — implementing the component foresight protocol of the SES system.

4.6 Novel Contribution 5 — OSERDESE3 8:1 DDR Serializer Configuration

The triton_pulse_heart module instantiates an OSERDESE3 primitive with the following configuration:

- **DATA_RATE_OQ:** DDR — both rising and falling edges of CLK are used for serialization
- **DATA_WIDTH:** 8 — 8 parallel input bits serialized per CLKDIV cycle
- **CLK:** clk_fast (600 MHz) — from BUFG(clk_fast_unbuf)
- **CLKDIV:** clk_div (150 MHz) — from BUFGCE_DIV(clk_fast_unbuf ÷ 4)
- **D[7:0]:** 8-bit parallel pattern from frequency_pattern_encoder
- **OQ:** 1-bit serial output to OBUFDS

A 3-stage flip-flop synchronizer in the clk_div domain safely crosses the active-low reset signal from the clk_100m domain, preventing metastability at the OSERDESE3 RST input. The OBUFDS drives pulse_out_p and pulse_out_n with DIFF_HSTL_I_12 I/O standard, appropriate for UltraScale+ High-Performance I/O banks operating at 1.2V VCCO (Bank 64 on Alveo U250).

4.7 Novel Contribution 6 — CRC64-ECMA-182 Hard Delivery Gate

The crc64_seal module computes a 64-bit CRC over each pulse frame using the CRC64-ECMA-182 polynomial (0x42F0E1EBA9EA3693). The seal_intact signal asserted by the module is a combinational prerequisite in the Conductor FSM for lane promotion. The FSM transition from ST_SEAL_WAIT to ST_PROMOTE has the condition:

next_state <= ST_PROMOTE when seal_intact = '1' else ST_SEAL_WAIT;

There is no timeout, bypass, or software override. CRC32 (collision probability ~1:4 billion) is architecturally rejected: in a frequency-domain encoding system generating millions of frames per second, CRC32 collision probability over a multi-hour run is statistically non-negligible. CRC64-ECMA-182 (collision probability ~1:18 quintillion) is the minimum acceptable integrity mechanism.

4.8 Novel Contribution 7 — Conductor FSM Lane Arbitration

The conductor_fsm module is the central control plane of the Shimmers oscillator engine. It orchestrates: sync_pulse issuance to gate phase accumulator advancement; chamber_select control for dual-chamber URAM alternation; crc_valid monitoring for integrity gate enforcement; spoke_ready gating for lane promotion; chamber_switch issuance for stack_reader drain triggering; and crc_msg_end/crc_start for frame boundary delineation.

The FSM ensures all pulse delivery is deterministic: no stalls are permitted in the path from phase accumulator to pulse output. All clock domain crossings use two-stage flip-flop synchronizers. The xadc_alert input from the Sysmon loop enables autonomous lane switching in response to thermal or voltage anomalies without system halt.

4.9 Novel Contribution 8 — Sysmon Real-Time Frequency Compensation Loop

The xadc_monitor module wraps the XADC (7-series) or Sysmon (UltraScale+) hard IP block to continuously measure on-chip junction temperature and core supply voltage:

$$T\ (^\circ C) = (ADC\_CODE \times 503.975 / 4096) - 273.15$$

$$V\ (V) = ADC\_CODE \times 3.0 / 4096$$

These measurements are converted to instruction words that feed an oscillator modifier pipeline, enabling real-time compensation of frequency drift caused by thermal load or supply voltage variation. No system halt, no bitstream reload, no host CPU intervention. The xadc_alert signal is additionally presented to the Conductor FSM to enable autonomous lane switching if anomalies exceed configured thresholds. Sysmon is a telemetry-and-compensation loop, not a shutdown handler.

4.10 Integration as Part of the SES Compiled Architecture

When Shimmers operates as part of the Shimmers · Eidolon · SPECTRE (SES) integrated system:

- **Shimmers feeds Eidolon:** The OSERDESE3 pulse output stream is the data source for Eidolon's Inception tier. Each frequency-encoded pulse frame is handed off via the Dragoncache hub interface after CRC64 seal assertion by the Conductor FSM.

- **Shimmers shares the frequency vocabulary with SPECTRE:** The MIDI-mapped bijective LUT in Shimmers is the same 128-frequency vocabulary that SPECTRE's Goertzel decoder uses for data recovery. The two components are designed around the same frequency alphabet, making the end-to-end system a coherent lossless pipeline.

- **Emergent system property:** In the SES system, the Shimmers oscillator generates the primary frequency events that drive the entire pipeline. The component foresight protocol (Eidolon pre-broadcasting pulse frames before delivery) means SPECTRE's Goertzel window begins accumulating samples from Shimmers before the formal delivery event — absorbing window-fill latency into the broadcast phase.

CLAIMS

(Informal claims for provisional filing — formal claims to be developed with patent attorney for non-provisional application)

Claim 1. A digital pulse generation system comprising: a Mixed-Mode Clock Manager generating a first clock signal at a first frequency from a differential reference input; a clock divider primitive driven from a pre-buffer version of the first clock signal and generating a second clock signal at a second frequency equal to the first frequency divided by an integer factor, wherein the first and second clock signals share an identical clock tree distribution root; an output serializer primitive having a first clock input connected to the first clock signal and a second clock input connected to the second clock signal; a phase accumulator advancing by a programmable phase increment each cycle of a fabric clock; a pattern encoder translating phase accumulator values to parallel data patterns; and a differential output buffer converting the serialized output of the serializer primitive to a differential pulse stream.

Claim 2. The system of Claim 1, wherein the clock divider primitive is a BUFGCE_DIV with a divide factor of 4, and wherein the first frequency is approximately 600 MHz and the second frequency is approximately 150 MHz, and wherein the Mixed-Mode Clock Manager VCO frequency is 1,200 MHz within the device-specified valid VCO range.

Claim 3. The system of Claim 1, further comprising a frequency lookup table implementing a bijective mapping from a 7-bit input index to a 32-bit phase increment word encoding a target frequency as a fraction of the fabric clock frequency, wherein the 7-bit input index corresponds to a MIDI note value in the range 0–127.

Claim 4. The system of Claim 3, wherein distinct input values map to distinct output frequencies across the full 128-entry input range, enabling lossless recovery of the original input symbol by a downstream frequency detector.

Claim 5. The system of Claim 1, further comprising a Huffman entropy encoder processing input bytes prior to frequency lookup, wherein the encoder enforces a most-significant-bit-zero invariant on all output bytes, guaranteeing that all output values are valid frequency table indices.

Claim 6. The system of Claim 1, further comprising a CRC64-ECMA-182 integrity seal module computing a 64-bit cyclic redundancy check over each pulse frame, wherein delivery of said pulse frame to a downstream consumer is conditioned on assertion of a seal_intact signal from the CRC module, with no bypass or timeout.

Claim 7. The system of Claim 6, wherein a Conductor finite state machine arbitrates lane promotion, chamber switching, and sync pulse issuance, and wherein the transition from a seal-wait state to a promote state is conditioned solely on the seal_intact signal, with no software override.

Claim 8. The system of Claim 1, further comprising a sensor monitoring subsystem continuously measuring on-chip junction temperature and core supply voltage and generating compensation signals to regulate oscillator frequency in real-time response to detected anomalies, without system halt or bitstream reload.

Claim 9. The system of Claim 1, implemented on a Field-Programmable Gate Array device and achieving sustained pulse output at or above 600 million pulses per second with static timing analysis confirming positive worst-case setup slack (WNS > 0), positive worst-case hold slack (WHS > 0), and positive worst-case pulse width slack (WPWS > 0) simultaneously.

Claim 10. A method of generating high-frequency digital pulses comprising: generating a high-speed clock and a divided clock from the same pre-buffer clock source to guarantee identical distribution path to a serializer primitive; mapping an input data symbol to a phase increment word via a bijective lookup table indexed by MIDI note values 0–127; advancing a phase accumulator by the phase increment word each fabric clock cycle; encoding the instantaneous phase value as an 8-bit parallel pattern; serializing the pattern via a hardware output serializer at the high-speed clock rate to produce a pulse stream at or above 600 MHz; computing a 64-bit CRC over the pulse frame; and conditioning delivery of the pulse frame on successful CRC64 verification.

Claim 11. The method of Claim 10, further comprising monitoring on-chip temperature and voltage and dynamically adjusting oscillator frequency to compensate for environmental variation without system halt or bitstream reload.

Claim 12. The method of Claim 10, further comprising entropy-encoding input bytes prior to frequency lookup using a Huffman encoder that enforces a most-significant-bit-zero constraint on all output codes, guaranteeing frequency table index validity before lookup.

ABSTRACT

Shimmers is a high-frequency digital pulse generation system that overcomes the fundamental limitations of prior art Direct Digital Synthesis by replacing ROM-based waveform lookup with a hardware-native OSERDESE3 output serializer driven by a Mixed-Mode Clock Manager. A novel clock generation topology derives the serializer's divided control clock (CLKDIV) from the identical pre-buffer source as the high-speed clock (CLK) via a BUFGCE_DIV primitive, ensuring both signals share an identical clock tree root and eliminating the inter-tree routing skew that causes OSERDESE3 Max Skew violations in prior art implementations. Input data bytes are Huffman-encoded with a forced MSB=0 invariant guaranteeing MIDI note validity, then mapped to frequencies via a bijective 128-entry lookup table and a 32-bit phase accumulator, encoded as 8-bit parallel patterns, serialized at the full 600 MHz clock rate, and delivered as a differential pulse stream. CRC64-ECMA-182 integrity sealing is enforced as a hard prerequisite for pulse frame delivery by the Conductor FSM. A sensor monitoring loop provides real-time thermal and voltage compensation without system interrupt. The system is implemented in VHDL-2008 and has been validated with zero timing violations across nine distinct FPGA device families including Spartan-7 through Versal VPK180 and AMD Alveo datacenter accelerator cards.

INVENTOR DECLARATION

I hereby declare that I am the sole inventor of the subject matter claimed in this provisional patent application; that I believe I am the original and first inventor of the invention; that I have reviewed and understand the contents of this application; and that I acknowledge my duty to disclose information material to patentability.

All artificial intelligence tools used during the development and documentation of this invention were used as instruments under my direction. They made no independent inventive contribution. The inventive concept, architectural decisions, performance criteria, validation methodology, and reduction to practice were conceived and directed solely by me.

Inventor: Scott Slater **Date:** May 4, 2026

This provisional patent application establishes a priority date for the invention described herein. A non-provisional patent application claiming the benefit of this provisional must be filed within twelve (12) months of the filing date of this provisional pursuant to 35 U.S.C. § 119(e).

Filed under: SES-001-SHIMMERS | Docket: Shimmers · Eidolon · SPECTRE System | Priority: April 25, 2026 (parent project SPECTRE)